The documentation and process conversion measures necessary to comply with this revision shall be completed by 11 July 2014.

INCH-POUND

MIL-PRF-19500/540F
11 April 2014
SUPERSEDING
MIL-PRF-19500/540E
10 September 2010

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, DARLINGTON TRANSISTOR, PNP SILICON, POWER,
TYPES 2N6298 AND 2N6299, JAN, JANTX, AND JANTXV

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for PNP, Darlington, silicon, power transistors. Three levels of product assurance is provided for each device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figure 1 (similar to TO-66).

* 1.3 Maximum ratings. Unless otherwise specified, T_C = +25°C.

Type	P_T (1)		$R_{\theta JC}$	V_{CBO}	V_{CEO}	V_{EBO}	I_C	I_B	T_J and T_{STG}
	T_C = 25°C	T_C = +100°C							
	W	W	°C/W	V dc	V dc	V dc	A dc	mA dc	°C
2N6298	64	32	2.33	-60	-60	-5	-8	-120	-65 to +175
2N6299	64	32	2.33	-80	-80	-5	-8	-120	-65 to +175

(1) Derate linearly at 0.428 W/°C above T_C > 25°C.

* 1.4 Primary electrical characteristics. Unless otherwise specified, T_C = +25°C.

Limits	h_{FE2} (1)	h_{FE3} (1)	$\lvert h_{fe} \rvert$	C_{obo}	Pulse response		$V_{BE(sat)}$ (1)	$V_{CE(sat)2}$ (1)	h_{fe}
	V_{CE} = -3 V dc I_C = -4 A dc	V_{CE} = -3 V dc I_C = -8 A dc	V_{CE} = -3 V dc I_C = -3 A dc f = 1 MHz	100 kHz ≤ f ≤ 1 MHz V_{CB} = -10 V dc I_E = 0	t_{on}	t_{off}	I_C = -8 A dc I_B = -80 mA dc	I_C = -8 A dc I_B = -80 mA dc	V_{CE} = -3 V dc I_C = -3 A dc f = 1 kHz
				pF	µs	µs	V dc	V dc	
Min	750	100	25						300
Max	18,000		350	200	2.0	8.0	-4.0	-3.0	

(1) Pulsed (see 4.5.1).

* Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, ATTN: VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

AMSC N/A

FSC 5961

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2. APPLICABLE DOCUMENTS

* 2.1 General. The documents listed in this section are specified in sections 3 or 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 or 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

* (Copies of these documents are available online at http://quicksearch.dla.mil/).

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturers list before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface and physical dimensions. Interface and physical dimensions shall be as specified in MIL-PRF-19500, and on figure 1, (similar to TO-66).

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500.

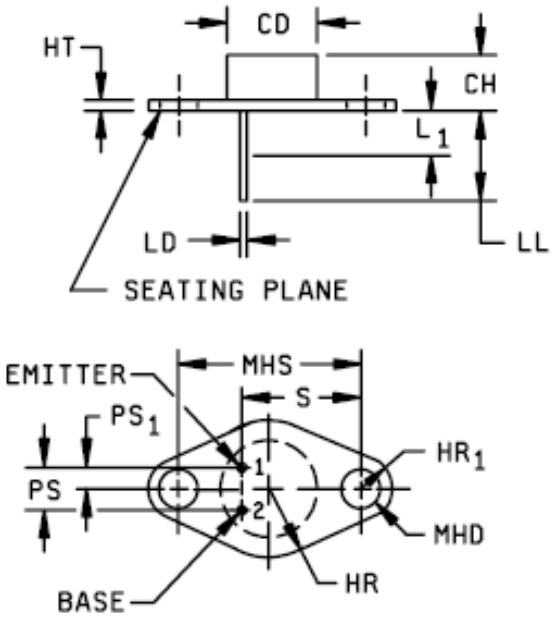
3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

3.7 Electrical test requirements. The electrical test requirements shall be as specified in table I.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

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Dimensions					
Symbol	Inches		Millimeters		Notes
	Min	Max	Min	Max	
CD		.620		15.76	
CH	.250	.340	6.35	8.64	
HR		.350		8.89	
HT	.050	.075	1.27	1.91	
HR_1	.115	.145	2.92	3.68	4
LD	.028	.034	.71	.86	4, 6
LL	.360	.500	9.14	12.70	
L_1		.050		1.27	6
MHD	.142	.152	3.61	3.86	4
MHS	.958	.962	24.33	24.43	
PS	.190	.210	4.83	5.33	3
PS_1	.093	.107	2.36	2.73	3
S	.570	.590	14.48	14.99	
Notes	1, 2, 5, 7				



NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. These dimensions should be measured at points .050 inch (1.27 mm) +.005 inch (0.13 mm) -.000 inch (0.00 mm) below seating plane. When gauge is not used, measurement will be made at the seating plane.
4. Two places.
5. The seating plane of the header shall be flat within .001 inch (0.03 mm) concave to .004 inch (0.10 mm) convex inside a .930 inch (23.62 mm) diameter circle on the center of the header and flat within .001 inch (0.03 mm) concave to .006 inch (0.15 mm) convex overall.
6. Lead diameter shall not exceed twice LD within L_1.
7. Lead number 1 is the emitter, lead 2 is the base, case is the collector.
8. In accordance with ASME Y14.5M, diameters are equivalent to ∅x symbology.

FIGURE 1. Physical dimensions (similar to TO-66).

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4 and tables I and II).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table II tests, the tests specified in table II herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

* 4.3 Screening (JANTX and JANTXV levels only). Screening shall be in accordance with table E-IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table E-IV of MIL-PRF-19500)	Measurements JANTX and JANTXV levels only
(1) 3c	Thermal impedance (see 4.3.2).
9	I_{CEX1} and h_{FE2}
11	I_{CEX1} and h_{FE2}
12	See 4.3.1
13	Subgroup 2 of table I herein; ΔI_{CEX1} = 100 percent of initial value or -2 μA dc, whichever is greater. Δh_{FE2} = ±40 percent.

(1) Shall be performed anytime after temperature cycling, screen 3a; and does not need to be repeated in screening requirements.

* 4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: $V_{CE} \geq$ -10 V dc; T_J = +162.5°C ±12.5°C.
NOTE: No heat sink or forced air cooling on the devices shall be permitted.

4.3.2 Thermal impedance. The thermal impedance measurements shall be performed in accordance with method 3131 of MIL-STD-750 using the guidelines in that method for determining I_M, I_H, t_H, t_{SW} (and V_H where appropriate). See table II, group E, subgroup 4 herein. Measurement delay time (t_{MD}) = 70 μs max.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with table E-V of MIL-PRF-19500 and table I herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

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* 4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VIB (JAN, JANTX, and JANTXV) of MIL-PRF-19500. Electrical measurements (end-points) shall be in accordance with the table I, subgroup 2 herein.

	Subgroup	Method	Condition
*	B3	1037	$V_{CE} \geq$ -10 V dc; ΔT_J between cycles $\geq$ +100°C; $t_{on} = t_{off}$ = 3 minutes for 2,000 cycles. No heat sink or forced-air cooling on the devices shall be permitted.

* 4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-VII of MIL-PRF-19500. Electrical measurements (end-points) shall be in accordance with the table I, subgroup 2 herein.

	Subgroup	Method	Condition
	C2	2036	Test condition A, weight = 10 pounds, time = 15 seconds.
	C5	3151	$R_{\theta JC}$ = 2.33°C/W.
*	C6	1037	$V_{CE} \geq$ -10 V dc; ΔT_J between cycles $\geq$ +100°C; $t_{on} = t_{off}$ = 3 minutes for 6,000 cycles. No heat sink or forced-air cooling on device shall be permitted.

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-IX of MIL-PRF-19500 and as specified herein. Electrical measurements (end-points) shall be in accordance with table I, subgroup 2 herein.

4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

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* TABLE I. Group A inspection.

	Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
	Subgroup 1						
	Visual and mechanical examination	2071					
	Subgroup 2						
	Thermal impedance 2/	3131	See 4.3.2	$Z_{\theta JC}$			°C/W
	Breakdown voltage, collector to emitter	3011	Bias condition D, I_C = -100 mA dc; pulsed (see 4.5.1)	$V_{(BR)CEO}$			
	2N6298				-60		V dc
	2N6299				-80		V dc
	Collector to emitter cutoff current	3041	Bias condition A, V_{BE} = -1.5 V dc	I_{CEX1}			
	2N6298		V_{CE} = -60 V dc			-10	µA dc
	2N6299		V_{CE} = -80 V dc			-10	µA dc
	Collector to emitter cutoff current	3041	Bias condition D	I_{CEO}			
	2N6298		V_{CE} = -30 V dc			-0.5	mA dc
	2N6299		V_{CE} = -40 V dc			-0.5	mA dc
	Emitter to base cutoff current	3061	Bias condition D, V_{EB} = -5 V dc	I_{EBO}		-2.0	mA dc
*	Forward-current transfer ratio	3076	V_{CE} = -3 V dc, I_C = -1 A dc, pulsed (see 4.5.1)	h_{FE1}	500		
*	Forward-current transfer ratio	3076	V_{CE} = -3 V dc, I_C = -4 A dc, pulsed (see 4.5.1)	h_{FE2}	750	18000	
*	Forward-current transfer ratio	3076	V_{CE} = -3 V dc, I_C = -8 A dc, pulsed (see 4.5.1)	h_{FE3}	100		
	Base emitter voltage (nonsaturated)	3066	Test condition B, V_{CE} = -3 V dc, I_C = -4 A dc, pulsed (see 4.5.1)	$V_{BE(on)}$		-2.8	V dc
	Base emitter voltage (saturated)	3066	Test condition A, I_C = -8 A dc, I_B = -80 mA dc, pulsed (see 4.5.1)	$V_{BE(sat)1}$		-4.0	V dc
	Collector - emitter voltage (saturated)	3071	I_C = -4 A dc, I_B = -16 mA dc, pulsed (see 4.5.1)	$V_{CE(sat)1}$		-2.0	V dc
	Collector - emitter voltage (saturated)	3071	I_C = -8 A dc, I_B = 80 mA dc, pulsed (see 4.5.1)	$V_{CE(sat)2}$		-2.0	V dc

See footnotes at end of table.

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* TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 3						
High temperature operation:		T_A = +150°C				
Collector to emitter cutoff current	3041	Bias condition A, V_{BE} = -1.5 V dc	I_{CEX2}			
2N6298		V_{CE} = -60 V dc			-5.0	mA dc
2N6299		V_{CE} = -80 V dc			-5.0	mA dc
Low temperature operation:		T_A = -55°C				
Forward current transfer ratio	3076	V_{CE} = -3 V dc, I_C = -4 A dc, pulsed (see 4.5.1)	h_{FE4}	200		
Subgroup 4						
Small signal short circuit forward current transfer ratio	3206	V_{CE} = -3 V dc; I_C = -3 A dc; f = 1 kHz	h_{fe}	300		
Magnitude of small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = -3 V dc, I_C = -3 A dc, f = 1.0 MHz	$\|h_{fe}\|$	25	350	
Pulse response	3251	Test condition B, except test circuit and pulse requirements in accordance with figure 2 for "t_{on}" and figure 3 for "t_{off}"				
Turn-on time		See figure 2; V_{CC} = -30 V dc; I_C = -4 A dc; I_{B1} = -16 mA dc	t_{on}		2.0	µs
Turn-off time		See figure 3; V_{CC} = -30 V dc; I_C = -4 A dc; I_{B1} = -16 mA dc	t_{off}		8.0	µs
Open circuit output capacitance	3236	V_{CB} = -10 V dc; I_E = 0; 100 kHz ≤ f ≤ 1 MHz	C_{obo}		200	pF

See footnotes at end of table.

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* TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 5						
Safe operating area (dc)	3051	$T_{C.}$ = 25°C +10°C, t = 1 s; 1 cycle; see figure 4;				
Test 1		V_{CE} = -8 V dc; I_C = -8 A dc				
Test 2		V_{CE} = -20 V dc; I_C = -2.0 A dc				
Test 3		I_C = -100 mA dc				
2N6298		V_{CE} = -60 V dc				
2N6299		V_{CE} = -80 V dc				
Safe operating area (switching)	3053	Load condition B, (clamped inductive load); T_A = +25°C; $t_r + t_f \leq 1.0$ µs duty cycle ≤ 10 percent, t_p = 1 ms; (vary to obtain I_C); R_S = 0.1 ohms; R_{BB1} = 80 ohms; V_{BB1} = -16 V dc; R_{BB2} = 100 ohms; V_{BB2} = -1.5 V dc; V_{CC} = -50 V dc; I_C = -8 A dc; $R_L \leq 2$ ohms; L = 1 mH; clamp voltage				
2N6298		-60 V dc				
2N6299		-80 V dc				
Safe operation area (switching)	3053	Load condition C; (unclamped inductive load) See figure 5, T_A = +25°C; duty cycle ≤ 10 percent R_S = 0.1 ohms				
Test 1		t_p = 1 ms; (vary to obtain I_C); R_{BB1} = 80 ohms; $V_{BB1} \geq$ -12 V dc; $R_{BB2} = \infty$; $V_{CC} \geq$ -30 V dc; I_C = -8 A dc; $R_L \leq$ 0.5 ohms; L = 1 mH at 8 A dc				

See footnotes at end of table.

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* TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 5 - Continued						
Test 2		t_p = 1 ms; (vary to obtain I_C); R_{BB1} = 80 ohms; $V_{BB1} \geq$ -12 V dc; $R_{BB2} = \infty$; V_{BB2} = 0 V dc; V_{CC} = -30 V dc; I_C = -0.2 A dc; $R_L \leq$ 0.5 ohms; L = 100 mH; at -0.2 A dc				
End point electrical measurements:						
Collector to emitter cutoff current	3041	Bias condition A; V_{BE} = -1.5 V dc	I_{CEX1}			
2N6298		V_{CE} = -60 V dc			-0.5	mA dc
2N6299		V_{CE} = -80 V dc			-0.5	mA dc
Forward-current transfer ratio	3076	V_{CE} = -3 V dc; I_C = -4 A dc; pulsed (see 4.5.1)	h_{FE2}	750	18,000	

1/ For sampling plan, see MIL-PRF-19500.
2/ This test required for the following end-point measurements only:
Group B, subgroups 2 and 3 (JAN, JANTX, and JANTXV).
Group C, subgroup 2 and 6.
Group E, subgroup 1.

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* TABLE II. Group E inspection (all quality levels) - for qualification and re-qualification only.

Inspection	MIL-STD-750		Sample plan
	Method	Conditions	
Subgroup 1			45 devices c = 0
Temperature cycling	1051	500 cycles.	
Hermetic seal	1071		
Fine leak Gross leak			
Electrical measurements		See table I, subgroup 2 herein.	
Subgroup 2			45 devices c = 0
Blocking life	1048	Test temperature = +125°C; V_{CB} = 80 percent rated; T = 1,000 hours.	
Electrical measurements		See table I, subgroup 2 herein.	
Subgroup 4			Sample size N/A
Thermal impedance curves		See table E-IX of MIL-PRF-19500, group E, subgroup 4.	
Subgroup 8			45 devices c = 0
Reverse stability	1033	Condition B.	

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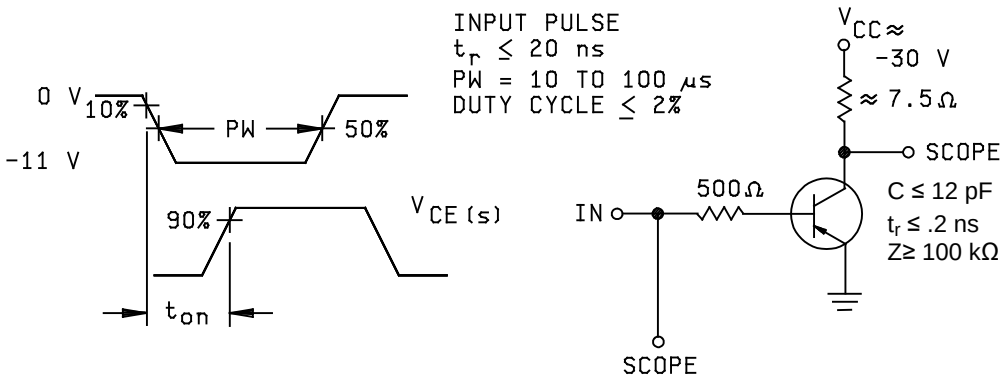


FIGURE 2. Turn-on time test circuit.

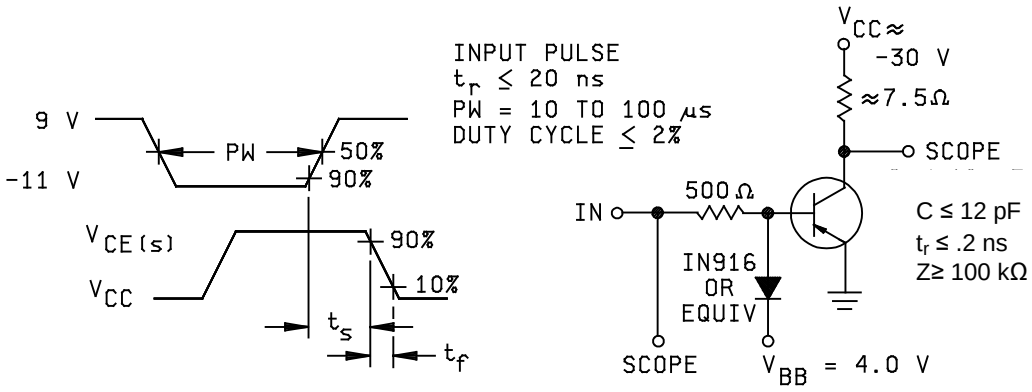


FIGURE 3. Turn-off time test circuit.

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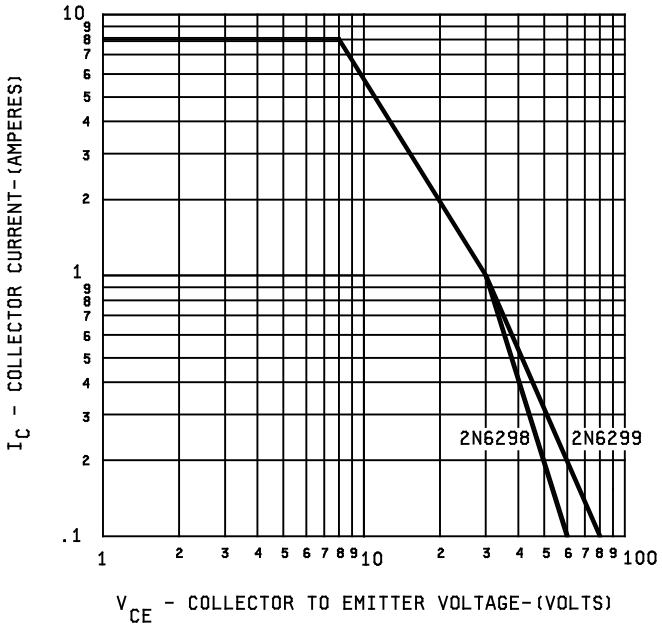


FIGURE 4. Maximum safe operating graph (dc).

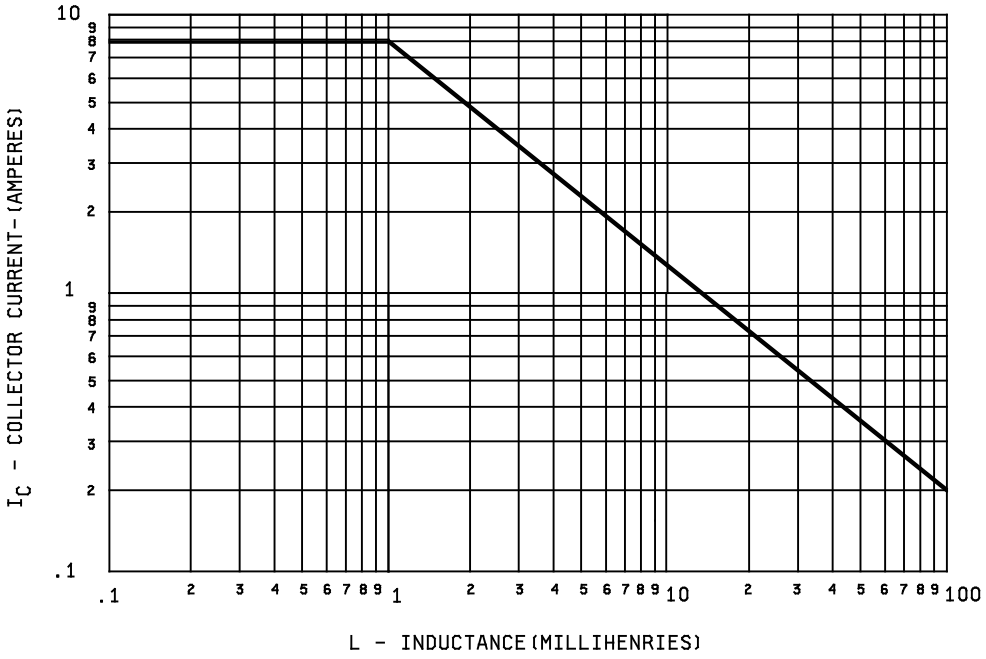


FIGURE 5. Safe operating area for switching between saturation and cutoff (unclamped inductive load).

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

* d. The complete Part or Identifying Number (PIN), see title and section 1.

* 6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML 19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil .

6.4 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Army - CR
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5961-2013-123)

Review activities:
Army - AR
Air Force - 19, 71, 99

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